

April 22, 2025

Dear Sir or Madam,

I am writing to commend Mahin Matlapudi and Riyad Babayev for their great work on the third project in the course I am teaching this semester, AE353 (Aerospace Control Systems).

The goal of this third project was to design, implement, and test in simulation a control system that enables a spacecraft with reaction wheels to maintain a fixed orientation so that an astronaut (i.e., a “space cat”) can safely dock and re-enter the spacecraft through a hatch. A key challenge was to estimate the full state of the spacecraft given noisy measurements from a star tracker.

Each pair of students submitted a 60-second video and a 6-page written report. Four (out of forty-seven) of the videos stood out to me. Mahin and Riyad’s video [link], in particular, told the “story of Neil Catstrong,” the “first feline astronaut.” The climax of their story was when, “against all odds, they found Neil floating in space, helmet cracked but purring strong—he came home a legend.” No, this story did not add to the technical content (which they also included in their video), but it sure was fun way to keep me engaged—I won’t soon forget it. For this reason, I chose to read their report in detail.

The most interesting part of this report to me was their choice of “actuator configuration” (where to put reaction wheels) and “sensor configuration” (which stars to track). Mahin and Riyad chose to put reaction wheels at the vertices of a tetrahedron “to enable full 3-axis control, reduce cross-coupling between axes, and provide redundancy.” They also chose to track stars “with wide angular separation to maximize observability of roll, pitch, and yaw.” I appreciated the explanation for these choices as well as the results—they achieved success in 80% of trials in experiment.

I am happy to be working with Mahin and Riyad this semester and encourage you to reach out to them if you are looking for talented engineers who know something about control systems.

Sincerely,



Timothy Bretl

Severns Faculty Scholar
Professor and Associate Head of Aerospace Engineering
Affiliate of the Coordinated Science Laboratory
University of Illinois at Urbana-Champaign
217-244-3126, tbretl@illinois.edu, <http://bretl.csl.illinois.edu/>